

EXAM, LINEAR ANALYSIS, MARCH 16 2022.

(Allowed tools: Pen or pencil, rubber.) The exam time is 5 hours. No cheating.

Remember to explain all non-trivial identities as clearly as possible.

PROBLEM 1

Find the radius of convergence of the following power series:

- a) $\sum_{n=1}^{\infty} n^4 x^n$
- b) $\sum_{n=5}^{\infty} (2 - 1/n)^n x^n$
- c) $\sum_{n=1}^{\infty} e^{n^2} x^n$

PROBLEM 2

Find a solution to the following problem:

$$\begin{cases} \partial_t u(x, t) = 2\partial_{xx}^2 u(x, t), & 0 < x < \pi, \ t > 0 \\ u(0, t) = u(\pi, t) = 0, & t > 0 \\ u(x, 0) = 1, & 0 < x < \pi \end{cases}$$

Remember to motivate well why you think your answer really solves the equation system!

PROBLEM 3

Let f be the 2π -periodic function given by $f(x) = x^2$ for $-\pi < x \leq \pi$.

- a) Find the Fourier series for f .
- b) Does the series converge uniformly to f ?
- c) Find the value of the series

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}.$$

(Hint: You may need to use that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.)

- d) Find the value of the series

$$\sum_{n=1}^{\infty} \frac{1}{n^4}.$$

PROBLEM 4

Find the polynomial $p(x)$ of degree ≤ 2 that minimizes

$$\int_0^1 |e^x - p(x)|^2 dx.$$

PROBLEM 5

Let f be the function given by

$$f(x) = \sum_{n=1}^{\infty} e^{-nx}, \quad x > 0.$$

Prove that

- a) the series converges uniformly on each compact subinterval of $(0, \infty)$.
- b) f is differentiable at each $x > 0$ and compute $f'(x)$.
- c) $\int_0^{\infty} f(x) dx = \infty$.

PROBLEM 6

Let a_n be the Fourier coefficients of the function in Problem 3, and let b_n be the Fourier coefficients of the 2π -periodic function which equals x^2 on $[0, 2\pi]$. Prove, without computing the b_n 's,

- a) that $(\frac{b_n}{a_n})_{n=1}^{\infty}$ is an unbounded sequence.
(Hint: You may use that $a_n = O(1/n^2)$ as $n \rightarrow \pm\infty$.)
- b) that $\sum_{n=-\infty}^{\infty} b_n = 2\pi^2$.

Also explain why it is natural to expect that the a_n 's go faster to 0 than the b_n 's, as $n \rightarrow \pm\infty$.